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Dear Editors,

Please consider the enclosed manuscript, *A Structural Spline-Penalized Tail Bound for L-Functions*, for publication.

What this paper is — and is not. This is *not* a proof of the Riemann Hypothesis, and it claims no equivalence with it. It introduces a finite-window, positive functional — the Spline–Penalized Tail Bound F_λ — and studies it as a *computable detector statistic* for off-critical zeros, built explicitly on top of classical oscillation theory. The forward direction (an off-critical zero forces exponential growth of F_λ) is, for the genuine explicit-formula remainder $G_\sigma(t) = e^{-\sigma t}(\psi(e^t) - e^t)$, a *read-out* of the Turán–Pintz Ω-oscillation phenomenon, not a new forcing mechanism; we say so plainly throughout.

What is proved. For a tractable square-summable surrogate H_σ of G_σ we prove (i) an essentially unconditional polynomial variance bound $F_\lambda = O(T \log T \log\log T)$ when all zeros lie on or left of σ, with explicit blockwise constants; and (ii) exponential growth $F_\lambda \gg e^{2(\beta-\sigma)T}$ under any off-critical zero. The converse (the "Horocycle Conjecture") is proved for finite and attained-edge zero configurations and reduced, in the remaining unattained-edge case, to a precise rigidity statement about non-clustering of ordinates. We are explicit about the two open inputs — a mild ordinate-spacing hypothesis (far weaker than RH or pair correlation) and the unattained-edge rigidity — and localize each exactly.

Numerics and reproducibility. The analytic dichotomy is exercised numerically on Odlyzko's zeros, with measured exponential slopes matching the theoretical value to within about 10^{-3} relative error; we are careful to label the bias experiments as calibration of the estimator rather than independent confirmation. Full `R` source, the zero data, and all tables/figures are released for reproduction.

Suggested venue. Given its character — a computable diagnostic with explicit constants, numerical validation, and honestly delimited open inputs, rather than a new analytic criterion for RH — the paper seems best suited to an experimental or computational venue such as *Experimental Mathematics* or *Mathematics of Computation*.

Thank you for your consideration.

Sincerely,
Akbar Akbari Esfahani